IGCSE 0607 Practice Questions — E4.3

Length and midpoint

Total Questions: 18 — **Total Marks:** 121 Marks

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1 The points A and B have coordinates $(2, -3)$ and $(8, 5)$ respectively.

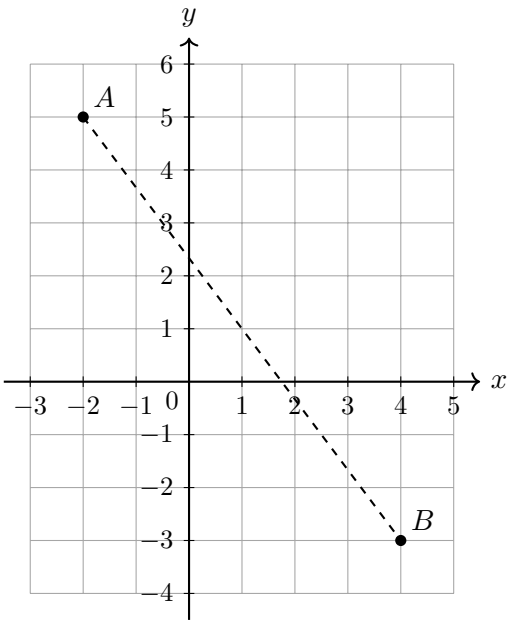
(a) Find the coordinates of the midpoint of AB.

(..... , ...) **[2]**

(b) Calculate the length of the line segment AB.

$AB =$ ..... **[2]**

2 The points $A(-2, 5)$ and $B(4, -3)$ are shown on the coordinate grid.



(a) Find the coordinates of the midpoint of the line segment AB.

((..... , ...)) **[2]**

(b) Calculate the exact length of the line segment AB.

..... **[3]**

3 Point A has coordinates $(1, 4)$ and point B has coordinates $(7, -4)$.

(a) Calculate the coordinates of the midpoint of the line segment AB.

(..... , ...) **[2]**

(b) Calculate the length of the line segment AB.

..... **[3]**

4 The points A and B have coordinates $(2, -3)$ and $(8, 5)$ respectively.

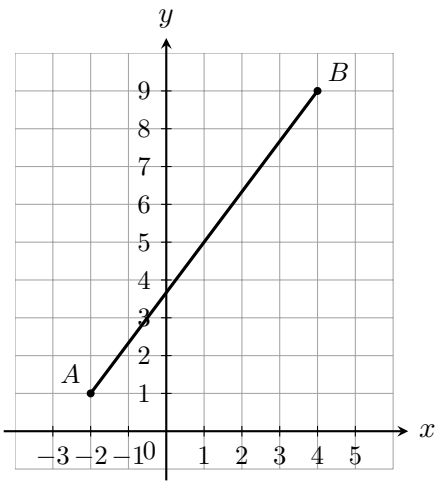
(a) Find the coordinates of the midpoint of the line segment AB.

(..... , ...) **[2]**

(b) Calculate the exact length of the line segment AB.

..... **[3]**

5 The diagram shows a line segment AB plotted on a Cartesian coordinate grid.



(a) Write down the coordinates of point A and point B.

$A =$ ..... , $B =$ ..... **[2]**

(b) Find the coordinates of the midpoint M of the line segment AB.

$M =$ (..... , ...) **[2]**

(c) Calculate the length of the line segment AB.

..... units **[2]**

6 The point $M(4, 1)$ is the midpoint of the line segment joining $P(-2, 7)$ and Q.

(a) Find the coordinates of Q.

(..... , ...) **[2]**

(b) The point R has coordinates $(k, 10)$, where k is a constant.
The length of PR is 5.
Find the two possible values of k.

$k =$ ..... , $k =$ ..... **[3]**

7 A circle has its centre at the origin, $O(0, 0)$.
The point $P(-3, 4)$ lies on the circumference of this circle.

(a) Work out the radius of the circle.

..... **[2]**

(b) The line segment PQ is a diameter of the circle.
Find the coordinates of Q.

(..... , ...) **[2]**

(c) The point R has coordinates $(k, \sqrt{11})$.
R also lies on the circumference of the circle, and k is a negative number.
Work out the exact value of k.

$k =$ ..... **[3]**

8 The line segment PQ has midpoint $M(3, -1)$.
The coordinates of point P are $(-2, 4)$.

(a) Find the coordinates of point Q.

$Q = (\ldots\ldots\ldots\ ,\ \ldots\ldots\ldots)$ **[2]**

(b) Calculate the exact length of PQ, giving your answer in the form $k\sqrt{2}$, where k is an integer.

$PQ = \ldots\ldots\ldots\ldots\ldots$ **[3]**

(c) Point R lies on the y-axis such that $PR = QR$.
Find the coordinates of R.

$R = (\ldots\ldots\ldots\ ,\ \ldots\ldots\ldots)$ **[4]**

9 The points $A(-2,-1)$, $B(4,3)$ and $C(8,-3)$ form a triangle.
M is the midpoint of AB and N is the midpoint of AC.

(a) Find the coordinates of M and the coordinates of N.

$M =$ ..... , $N =$ ..... **[2]**

(b) Show that the length of MN is exactly half the length of BC.

[4]

10 Points A and B have coordinates $(3p, p-2)$ and $(p, 2p+1)$ respectively, where p is a constant.

(a) The midpoint of AB lies on the line $y = x + 1$.
Find the value of p.

$p =$ ..... **[3]**

(b) Given instead that the length of the line segment AB is 6, show that $5p^2 + 6p - 27 = 0$.

[3]

(c) Solve the equation $5p^2 + 6p - 27 = 0$ to find the two possible values of p.

$p =$ ..... , $p =$ ..... **[3]**

11 A circle has diameter AB, where A is the point $(-1, 2)$ and B is the point $(5, 8)$.

(a) Find the coordinates of the centre of the circle, C.

(..... , ...) **[2]**

(b) Find the exact radius of the circle.
Give your answer in the form $a\sqrt{b}$, where a and b are positive integers and b is as small as possible.

..... **[3]**

(c) Point D is such that the midpoint of CD is $M\left(\frac{5}{2}, \frac{3}{2}\right)$.

Find the exact length of AD, giving your answer in the form $c\sqrt{d}$, where c and d are integers and d is as small as possible.

..... **[5]**

12 The points A and B have coordinates $(-2, 5)$ and $(4, -3)$ respectively.

(a) Calculate the exact length of the line segment AB.

..... **[3]**

(b) Find the coordinates of the midpoint of AB.

(..... , ...) **[2]**

13 Point P has coordinates $(3, k)$ and point Q has coordinates $(7, 5)$.
The length of the line segment PQ is 5 units.

(a) Find the two possible values of k.

$k =$ ..... , $k =$ ..... **[3]**

(b) M is the midpoint of PQ.
For the case where $k < 5$, find the coordinates of M.

((..... , ...)) **[2]**

14 The midpoint of the line segment PQ is $M(1, -2)$.
The coordinates of P are $(-3, 5)$.

(a) Find the coordinates of Q.

(..... , ...) **[2]**

(b) Calculate the length of the line segment PQ.
Give your answer correct to 3 significant figures.

..... **[3]**

15 Point P has coordinates $(-3, 5)$ and point Q has coordinates $(k, -1)$.
The length of the line segment PQ is 10 units.

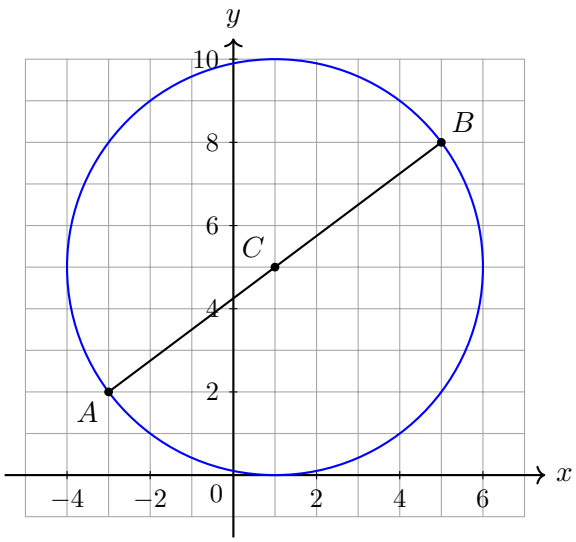
(a) Find the two possible values of k.

$k =$ ..... , $k =$ ..... **[4]**

(b) Given that $k > 0$, point M is the midpoint of PQ.
Find the coordinates of M.

(..... , ...) **[2]**

16 The points $A(-3, 2)$ and $B(5, 8)$ are the endpoints of a diameter of a circle.



(a) Find the coordinates of the centre of the circle, C.

((..... , ...)) **[2]**

(b) Calculate the area of the circle.
Give your answer correct to 1 decimal place.

..... **[3]**

(c) The point $D(k, 8)$ lies on the circumference of the circle.
Find the possible values of k.

$k =$ ..... , $k =$ ..... **[3]**

17 The points $A(2p, p)$ and $B(3p, 4p)$ are joined by a straight line, where $p > 0$.

(a) The length of the line segment AB is $5\sqrt{10}$.
Find the value of p.

$p =$ ..... **[4]**

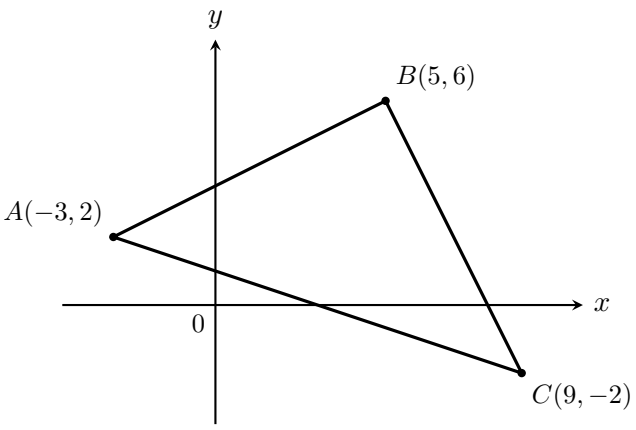
(b) Point C lies on the line segment AB such that the ratio $AC : CB = 2 : 3$.
Using your value of p from part (a), find the coordinates of C.

(..... , ...) **[3]**

(c) The point $D(x, y)$ is such that C is the midpoint of BD.
Work out the coordinates of D.

(..... , ...) **[2]**

18 The diagram shows a sketch of triangle ABC with vertices $A(-3, 2)$, $B(5, 6)$, and $C(9, -2)$.



(a) Show that triangle ABC is isosceles.

[4]

(b) M is the midpoint of AC.

(i) Find the coordinates of M.

$M = (\ldots\ldots\ ,\ \ldots\ldots)$ **[2]**

(ii) Calculate the length of BM, giving your answer correct to 3 significant figures.

$BM = \ldots\ldots\ldots\ldots\ldots$ **[2]**

(c) Calculate the area of triangle ABC.

..... square units **[2]**

(d) Point D is such that $ABCD$ is a square.
Find the coordinates of D.

$D = ($ ..... , ... $)$ **[2]**